Instrumental Variables

SW Chapter 12

ECON3500: Econometrics and Applications

Spring 2026

Learning Objectives

Understand **why** we need instrumental variables to address endogeneity

Identify the three key characteristics of a valid instrument

Explain the mechanics of **two-stage least squares** (2SLS)

Test for **weak instruments** using first-stage F-statistics

Combine IV with **panel data** and fixed effects

Textbook Coverage

- 12.1: IV estimator with single regressor and single instrument
- 12.2: General IV regression model
- 12.3: Checking instrument validity
- 12.4/12.5: Applied examples

Why Instrumental Variables?

The Endogeneity Problem

Three threats to internal validity all produce $E(u_i|X_i) \neq 0$:

Omitted variable bias: An unobserved variable correlated with X is left out

Simultaneous causality: X causes Y, but Y also causes X

Errors-in-variables bias: X is measured with error

Endogeneity

All three problems result in $E(u_i|X_i) \neq 0$ — a violation of the zero conditional mean assumption. When this happens, OLS is **biased and inconsistent**.

What We've Tried So Far

Solutions to endogeneity considered previously:

Difference-in-differences (Ch. 10)

- Requires a clear before/after treatment

Fixed effects (Ch. 10)

- Requires panel data
- Endogeneity source must be time-constant
- Regressors must not be time-constant

Today: **Instrumental variables (IV)**

The most widely-used method for addressing endogeneity

IV can eliminate bias when $E(u_i|X_i) \neq 0$ using an **instrumental variable** Z

Motivating Example: Wages and Schooling

$$\log(wage_i) = \beta_0 + \beta_1 schooling_i + \delta V_i + u_i$$

β_1 measures the returns to schooling

One omitted variable V : innate ability as a worker

- Innate ability positively affects *wages* ($\delta > 0$)
- Likely that innate ability is positively correlated with *schooling*: $\text{corr}(schooling, V) > 0$

OLS estimator of β_1 may have **omitted variable bias**

If this is the only omitted variable, bias is **positive**

- $\hat{\beta_1}$ overestimates the financial returns to schooling

Can We Fix This with Multiple Regression?

$$\log(wage_i) = \beta_0 + \beta_1 schooling_i + \delta V_i + u_i$$

How do we measure innate ability?

IQ tests may capture *some* part of ability; hard to get IQ data for large samples

IQ is not a perfect measure of innate ability in the workplace

- Example: IQ tests wouldn't measure social skills
- Note: you *should* include IQ if available

Since IQ tests are imperfect, *schooling* likely remains correlated with the omitted part of innate ability

Multiple regression doesn't fully solve this problem

Can We Fix This with Panel Data?

$$\log(wage_i) = \beta_0 + \beta_1 schooling_i + \delta V_i + u_i$$

Innate ability might be reasonably constant over a career

But *schooling* is also typically constant for adult workers

Adults who go back to school after working are a non-representative group

Panel data do not provide convincing variation in schooling over a worker's career

Fixed effects won't work here

> ⚠ **Bottom Line**
>
> Neither multiple regression nor panel data convincingly addresses the ability bias in estimating returns to schooling. We need a different approach.

Intuition First: A Simple Example

Effect of Studying on Grades

What is the effect on grades of studying an additional hour per day?

Y = GPA

X = study time (hours per day)

Would you expect the OLS estimator of β_1 to be unbiased? Why or why not?

Stinebrickner and Stinebrickner (2008), “The Causal Effect of Studying on Academic Performance,” *The B.E. Journal of Economic Analysis & Policy*

$n = 210$ freshmen at Berea College (Kentucky) in 2001

Y = first-semester GPA

X = average study hours per day (time use survey)

Roommates were **randomly assigned**

Z = 1 if roommate brought a video game, = 0 otherwise

Is the Video Game Instrument Valid?

Do you think Z_i (whether a roommate brought a video game) is a valid instrument?

Is the instrument **powerful**?

- If your roommate brought a video game, you might study less
- First stage: video game treatment reduces study hours by 0.67 hours/day (significant)

Is the instrument **exogenous**?

- Roommates were **randomly assigned** — so Z is uncorrelated with unobserved student ability
- Random assignment makes this a strong argument

Is the instrument **excludable**?

- Does having a roommate with a video game directly affect your GPA, other than through study time?
- Plausible: the video game mainly affects grades by reducing study hours

Results: OLS coefficient on study = 0.038; IV coefficient = **0.36** (much larger). Why? **Selection bias** — students who study more tend to be **weaker** students compensating with extra effort (**negative selection**). This negative correlation between study time and ability biases OLS downward. IV corrects this by using only the variation in study time driven by the random video-game assignment.

Classic Examples

Philip Wright and the Demand for Butter (1928)

Philip Wright, *The Tariff on Animals and Vegetable Oils*

- Appendix B: "The Method of Introducing External Factors"
- Estimated supply and demand elasticities for butter and flaxseed oil

Data: total annual U.S. butter consumption and price, 1912–1922

Naive estimation: use OLS

$$\ln(Q_i^{butter}) = \beta_0 + \beta_1 \ln(P_i^{butter}) + u_i$$

Problem: price and quantity are *jointly determined* by supply and demand

The Identification Problem

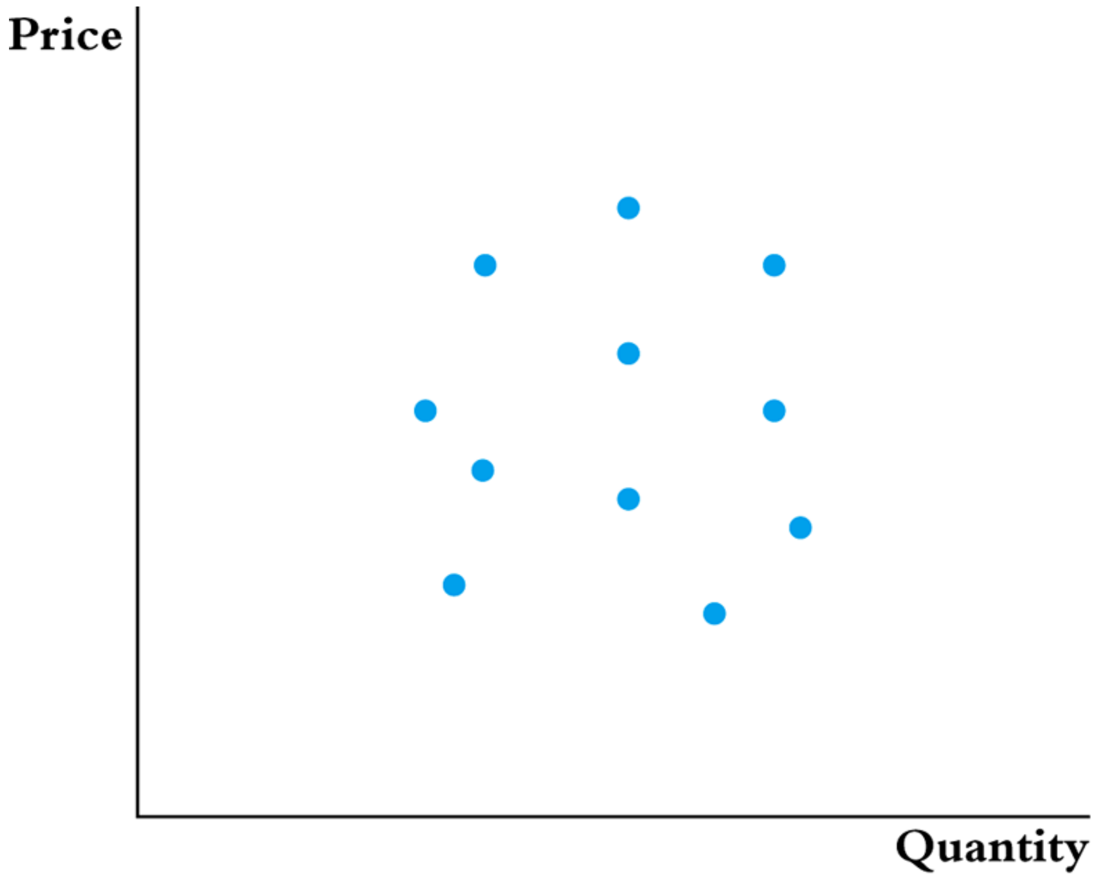
When both supply and demand shift, equilibrium data points trace out **neither** curve

We observe equilibrium prices and quantities

Can you tell what the demand curve looks like from these data points?

. . .

No — you just see a scatter of equilibrium points



(b) Equilibrium price and quantity for 11 time periods

A Better Way

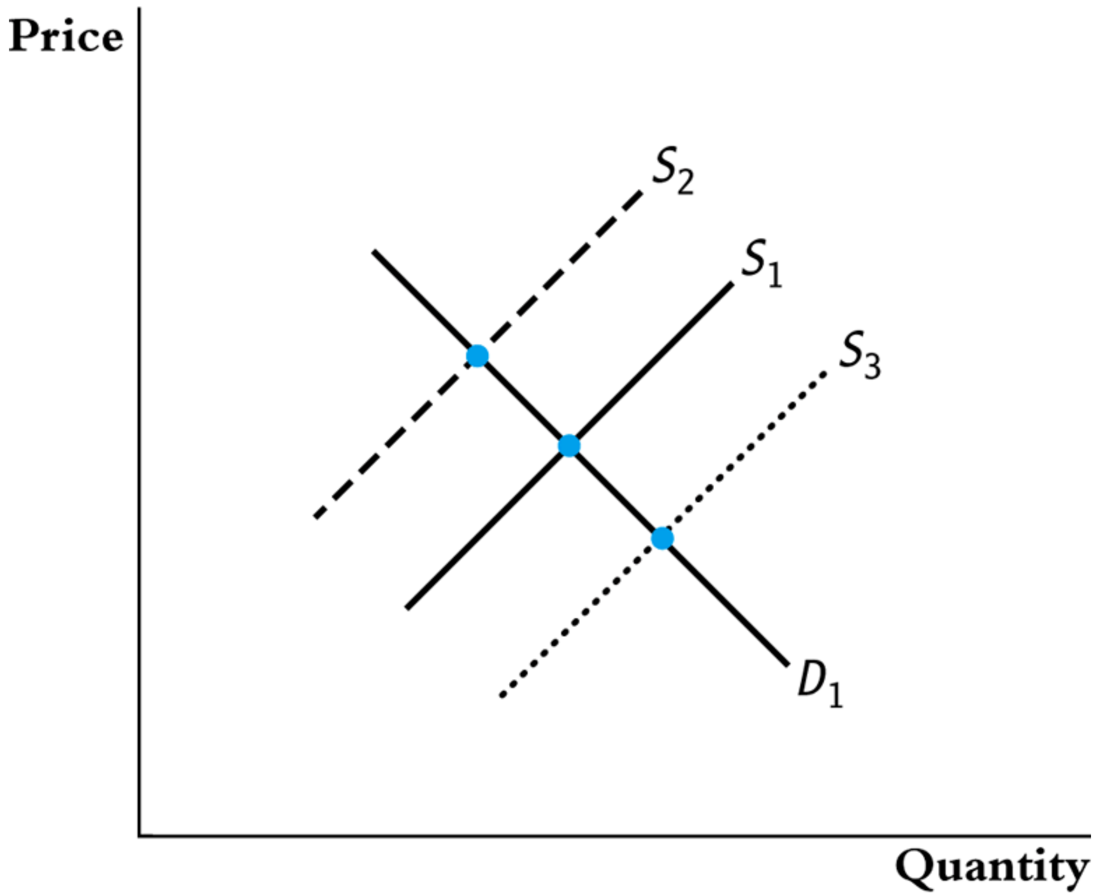
If you can **hold demand fixed** and only observe supply shifting:

The equilibrium points trace out the **demand curve**

This is the intuition behind IV

. . .

The **instrument** holds demand fixed while shifting supply — it moves X without directly affecting Y



(c) Equilibrium price and quantity when only the supply curve shifts

Demand for Cigarettes

Broad public policy interest in reducing cigarette consumption.

$$sales_i = 164.4 - 0.38 \cdot price_i + u_i$$

But price may be correlated with omitted variables in u

Prices in each state determined by cigarette firms

Firms may adjust price based on demand conditions

When state i has a high u_i, this state has unusually high demand

Therefore, $price_i$ may be positively correlated with u_i

> **Simultaneous Causality**
>
> Sales depend on prices, but prices may also depend on sales. This creates $\mathrm{Corr}(X, u) \neq 0$, making OLS inconsistent.

Why Simultaneous Causality Is Especially Problematic

$$sales_i = 164.4 - 0.38 \cdot price_i + u_i$$

Simultaneous causality is especially problematic because X_i will generally be correlated with **all** omitted variables in u_i

Hard to remove OVB by measuring the omitted variables

Would need to measure *every single* omitted variable

Simultaneous causality would *disappear* if we could **randomly assign** prices to the different states

In that experiment, there would be no correlation between *price* and u

IV provides a way to mimic this

The IV Solution

Instrumental Variables: Three Assumptions

An **instrumental variable** is an additional variable Z_i that satisfies three assumptions:

Z_i **is correlated** with X_i

- $\text{Corr}(Z, X) \neq 0$
- Z is a **powerful** or **relevant** instrument

Z_i **is not correlated** with the omitted variable, u_i

- $\text{Corr}(Z, u) = 0$
- Z is an **exogenous** instrument

Z_i **does not** *directly affect* (cause) Y_i

- It can only affect Y_i through its effect on X_i
- Z is an **excluded** instrument
- Z_i does not enter the equation $Y_i = \beta_0 + \beta_1 X_i + u_i$

> **Two vs. Three Assumptions**
>
> Some textbooks combine (2) and (3) into a single **validity** or **exogeneity** condition: $\text{Corr}(Z, u) = 0$. Others (including SW) separate them: **exogeneity** (Z uncorrelated with u) and **exclusion** (Z affects Y only through X). The practical question is the same either way: *can Z affect Y through any channel other than X?*

The Wald Estimator

With a **binary instrument** $Z_i \in \{0, 1\}$, there is a simple way to see how IV works:

First stage: How does Z move X?

$$\text{First stage} = E[X_i \mid Z_i = 1] - E[X_i \mid Z_i = 0]$$

Reduced form: How does Z move Y?

$$\text{Reduced form} = E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]$$

$$\boxed{\beta_1^{Wald} = \frac{\text{Reduced form}}{\text{First stage}}}$$

Intuition

The reduced form tells us how the instrument changes the outcome. The first stage tells us how the instrument changes treatment. Their ratio tells us the effect of treatment **induced by the instrument**.

Identification

Start with the population model:

$$Y = \beta_0 + \beta_1 X + u$$

Take the covariance of both sides with Z:

$$\mathrm{Cov}(Y, Z) = \mathrm{Cov}(\beta_0 + \beta_1 X + u, Z)$$

$$= \mathrm{Cov}(\beta_0, Z) + \mathrm{Cov}(\beta_1 X, Z) + \mathrm{Cov}(u, Z)$$

$$= 0 + \beta_1 \mathrm{Cov}(X, Z) + 0 \quad \text{by } \mathrm{Cov}(u, Z) = 0$$

Solving for β_1:

$$\boxed{\beta_1 = \frac{\mathrm{Cov}(Y, Z)}{\mathrm{Cov}(X, Z)}}$$

Which Assumptions Did We Use?

$$\beta_1 = \frac{\text{Cov}(Y, Z)}{\text{Cov}(X, Z)}$$

$\text{Cov}(Z_i, u_i) = 0$ — explicitly used in the derivation

$\text{Cov}(X_i, Z_i) \neq 0$ — used to divide by $\text{Cov}(X, Z)$; can't divide by zero!

Z_i does not affect Y_i directly — used to write the population model as $Y_i = \beta_0 + \beta_1 X_i + u_i$

Key Feature of IV

Notice we **never** assumed $\text{Cov}(X_i, u_i) = 0$. IV explicitly allows for endogeneity — that's the whole point!

Intuition for the IV Formula

$$\beta_1 = \frac{\text{Cov}(Y, Z)}{\text{Cov}(X, Z)}$$

Goal: estimate β_1, how X affects Y

Problem: We think X is correlated with u

Solution: Don't compare Y (which contains u) and X directly

- $\text{Cov}(X, Y)$ is explicitly *not* in our formula

Instead, see how Y moves with a third variable Z, and how X moves with Z

Z is exogenous: uncorrelated with u; Z also does not affect Y directly

If Y and X are both correlated with Z, the only explanation under our assumptions is that X causes Y according to β_1

Example Instrument: Distance to College

$$\log(wage_i) = \beta_0 + \beta_1 schooling_i + \delta V_i + u_i$$

Schooling and ability (V) are correlated — need an instrument for *schooling*

Distance from high school to nearest college:

- **Powerful**: Positively correlated with schooling attainment
- **Exogenous**: Uncorrelated with worker ability (V)
- **Excluded**: Growing up near a college doesn't directly *cause* higher wages

$$\beta_1 = \frac{\text{Cov}(\log \text{wage}, \text{distance})}{\text{Cov}(\text{schooling}, \text{distance})}$$

Denominator is positive (closer to college → more schooling)

Numerator is positive if people near a college earn higher wages as adults

Note: *not* because the distance causes the higher wage

$\text{Cov}(X, Y)$ does not appear — we do not compare someone's wage to their schooling

Two-Stage Least Squares

The 2SLS Estimator

For a dataset with n observations, replace population covariances with sample covariances:

$$\hat{\beta}_1^{2SLS} = \frac{\widehat{\mathrm{Cov}}(Y, Z)}{\widehat{\mathrm{Cov}}(X, Z)} = \frac{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{Y})(Z_i - \bar{Z})}{\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})(Z_i - \bar{Z})}$$

$$\hat{\beta}_0^{2SLS} = \bar{Y} - \hat{\beta}_1^{2SLS}\bar{X}$$

Called **two-stage least squares** — why will become apparent shortly.

What Are the Two Stages?

Stage 1: Regress X on Z (the "first stage")

$$X_i = \pi_0 + \pi_1 Z_i + v_i$$

Form the predicted values:

$$\hat{X_i} = \hat{\pi_0} + \hat{\pi_1} Z_i$$

Note: $X_i = \hat{X_i} + \hat{v_i}$

Stage 2: Regress Y on $\hat{X}$ (the "second stage")

$$Y_i = \beta_0 + \beta_1 \hat{X_i} + u_i$$

Intuition: Why 2SLS Works

First stage regresses X on Z

- Forms a "best guess" of X using only data on Z

The predicted $\hat{X}$ is **not** correlated with omitted variables in the second stage

- If we predict price using sales tax, predicted prices can't be correlated with unmeasured factors that affect demand
- We assumed exogeneity: sales tax is uncorrelated with omitted variables

Second stage regresses Y on $\hat{X}$

- $\hat{X}$ is "cleansed" of any correlation with omitted variables
- No more OVB or simultaneous causality bias

Key Insight

The instrument isolates the variation in X that is exogenous. 2SLS uses only this "clean" variation to estimate β_1.

The Local Average Treatment Effect

IV uses only the variation in X driven by the instrument — that's the whole point.

But this also means we can only observe the effect among people for whom the instrument **actually affects their treatment**.

Suppose a treatment improves *your* outcome by 2, but *my* outcome by only 1

And the instrument strongly affects whether *you* get treatment, but barely affects *me*

Then the IV estimate will be much closer to **2** than to 1

> ⓘ **Local Average Treatment Effect (LATE)**
>
> The IV estimate is **local** to the people whose treatment status is changed by the instrument, weighted by how much the instrument affects them. It is **not** the average effect for all units, or even for all treated units.

Better LATE Than Never?

Implication: The IV estimate may not represent the average effect for everyone — or even for those actually treated.

Compared to a randomized experiment, IV may be **less informative** about what would happen if treatment were expanded to a broader population

But IV still provides a **valid causal estimate** for the subgroup whose behavior is changed by the instrument

When exogenous variation is limited, LATE may be the best causal estimate available

Practical Implication

When interpreting IV results, ask: *For whom does the instrument induce treatment?* The IV estimate applies to that group — not necessarily to all units in the sample.

Cigarette Demand: IV in Practice

The Instrument: Sales Tax

$$sales_i = \beta_0 + \beta_1 \, price_i + u_i$$

Instrument for *price* of cigarettes? Need a Z_i that is:

- **Powerful**: Correlated with price
- **Exogenous**: Uncorrelated with u_i (unobserved demand factors)
- **Excluded**: Does not directly impact cigarette demand

Sales tax in state i?

- **Powerful**: Sales tax should be positively correlated with price (price measured inclusive of taxes)
- **Exogenous**: Plausible, but contestable: states with stronger anti-smoking preferences may both set higher taxes and have lower cigarette demand
- **Excluded**: Plausible, but contestable: the tax affects demand primarily through price, not through other channels

2SLS in Stata

Stata's `ivregress` command handles both stages automatically:

```
ivregress 2sls packpc (avgprs=tax), robust
```

Dependent variable: `packpc` (packs per capita)

Parentheses before `=`: endogenous regressor (`avgprs`)

After `=`: instrument (`tax`)

`robust`: heteroskedasticity-robust standard errors

Reading the Syntax

`ivregress 2sls Y (X=Z), robust`

Y = dependent variable (first after `2sls`)

X = endogenous regressor (before `=`)

Z = excluded instrument (after `=`)

Stata: 2SLS Results

```
. ivregress 2sls packpc (avgprs=tax), robust

Instrumental variables (2SLS) regression       Number of obs   =         96
                                                Wald chi2(1)    =      88.46
                                                Prob > chi2     =     0.0000
                                                R-squared       =     0.4219
                                                Root MSE        =     19.567

------------------------------------------------------------------------------
             |               Robust
      packpc | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      avgprs |  -.4208748   .0447474    -9.41   0.000    -.5085781   -.3331715
       _cons |    169.556   7.516482    22.56   0.000      154.824    184.2881
------------------------------------------------------------------------------
Instrumented:  avgprs
Instruments:   tax
```

2SLS estimates that a 1-unit increase in price **leads to** a decrease of 0.42 packs per capita.

Doing 2SLS Manually

To understand the two stages, we can do them by hand:

```
* Stage 1: Regress X on Z
regress avgprs tax, robust
predict avgprs_predict           // Save predicted values

* Stage 2: Regress Y on predicted X
regress packpc avgprs_predict, robust
```

The **point estimates** are identical to `ivregress`

But the **standard errors are wrong** in the manual approach!

- Manual first stage has sampling errors
- Stata doesn't know the predicted prices are *generated* regressors

> ⚠ **Always Use `ivregress`**
>
> The `ivregress` command uses the correct standard error formula that accounts for first-stage estimation. Never report standard errors from manual 2SLS.

Viewing Both Stages with `first`

```
ivregress 2sls packpc (avgprs=tax), robust first
```

The `first` option displays the first-stage regression alongside the second stage.

First stage (regress avgprs on tax):

Coefficient on `tax`: 2.45 (t = 19.78)

$F(1, 94) = 391.06$

This is a **strong** first stage

Second stage (regress packpc on $\hat{X}$):

Coefficient on avgprs: -0.421

Same as the 2SLS results

Reporting the First Stage

First stage shows how X and Z are related

It provides a **statistical test** of the relevance assumption: $\text{Corr}(X, Z) \neq 0$

Rule of Thumb: F > 10

A common rule of thumb is that the first-stage F-statistic should be **greater than 10**. If so, instruments are likely **powerful** (relevant). This threshold is a guide, not a bright line.

Checking Instrument Validity

Weak Instruments

What if the first-stage F-test is less than 10?

May have a **weak instrument**

Sample covariance of X and Z may be close to 0

$$\hat{\beta}_1^{2SLS} = \frac{\widehat{\text{Cov}}(Y, Z)}{\widehat{\text{Cov}}(X, Z)}$$

Intuition: dividing by something close to zero **blows up** your estimate

Large standard errors and unreliable point estimates

> ⚠️ **Consequences of Weak Instruments**
>
> If the first-stage F-statistic is less than 10, the 2SLS estimates may be biased and confidence intervals may have incorrect coverage. Do not trust the results.

Which Assumptions Can Be Tested?

Assumption	Testable?	How?
Relevance ($\text{Corr}(Z, X) \neq 0$)	Yes	First-stage F-test (rule of thumb: > 10)
Exogeneity ($\text{Corr}(Z, u) = 0$)	No	Must argue based on institutional knowledge
Exclusion (Z doesn't directly cause Y)	No	Must argue based on theory

Exogeneity Cannot Be Tested

You lack data on the omitted variable, so you cannot test whether Z is correlated with it. Exogeneity must be **defended with reasoning** about the instrument and the omitted variables in question.

Knowledge Check

For the cigarette example, why might someone challenge the exogeneity of sales tax as an instrument? Could states with stronger anti-smoking attitudes set both higher taxes *and* have lower demand?

IV with Multiple Regressors

Adding Exogenous Controls

$$sales_i = \beta_0 + \beta_1 price_i + \beta_2 income_i + u_i$$

Income per person at the state level may affect cigarette sales

Income is **not** determined simultaneously with cigarette demand

- We assume income is uncorrelated with the composite error u

2SLS can handle variables **not treated as endogenous**

Income enters **both** stages:

- First stage: helps predict price
- Second stage: controls for income's direct effect on sales

```
ivregress 2sls packpc (avgprs=tax) incomepop, robust first
```

`incomepop` appears **outside** the parentheses → exogenous control

`avgprs` is inside parentheses → endogenous regressor

`tax` is after `=` → excluded instrument

Need an *Excluded* Instrument

We need at least one instrument for **each** regressor treated as endogenous in the outcome equation

Even if we have income as a regressor, we still need `tax` as the excluded instrument

Stata will give you an error message if there is no excluded instrument

> **Terminology**
>
> - **Included instruments**: Exogenous regressors (like income) that appear in both stages
> - **Excluded instruments**: Variables (like sales tax) that appear in the first stage but *not* the second stage
> - The "excluded" instrument is what gives IV its identifying power

Panel Data, Fixed Effects, and IV

Combining IV with Panel Data

$$sales_{it} = \alpha_i + \lambda_t + \beta_1 \, price_{it} + \beta_2 income_{it} + \delta V_i + \omega_{it}$$

Instrument for *price* is *sales tax*

Panel data with fixed effects **and** instrumental variables; data from 1985 & 1995

State fixed effects (α_i): Control for time-invariant omitted factors (e.g., a state's attitude towards smoking)

Time fixed effects (λ_t): Control for factors affecting all states in one year (e.g., national anti-smoking campaigns)

Instrument (sales tax): Addresses simultaneous causality between demand factors and price

Income is assumed exogenous

```
xtivreg packpc (avgprs = taxs) incomepop y1995, fe vce(cluster state)
```

Building Up: No Instruments

Step 1: State FE only

```
xtreg packpc avgprs incomepop, vce(cluster state) fe
```

Price coefficient: -0.355

Step 2: State + Year FE

```
xtreg packpc avgprs incomepop i.year, vce(cluster state) fe
```

Price coefficient: -0.416

Both estimates still suffer from simultaneous causality bias — price responds to state and time demand shocks.

Adding the Instrument

Step 3: State + Year FE with IV

```
xtivreg packpc (avgprs = taxs) incomepop y1995, vce(cluster state) fe first
```

First stage (predict avgprs using taxs, incomepop, y1995, state FE):

Coefficient on `taxs`: 1.17 (t = 19.04)

$F(3, 47) = 3001.56$ — **very strong** first stage

Second stage:

Price coefficient: -0.409 ($p < 0.001$)

Income: -1.68 ($p = 0.048$)

Using Logarithms for Elasticities

$$\log(sales_{it}) = \alpha_i + \lambda_t + \beta_1 \log(price_{it}) + \beta_2 \log(income_{it}) + \delta V_i + \omega_{it}$$

Instrument for log(price) is log(sales tax)

Using logs lets us interpret coefficients as elasticities, and with year fixed effects it also helps compare proportional changes over time

Coefficient on log(price) is the **elasticity** of sales w.r.t. price

```
xtivreg lpackpc (lavgprs = ltaxs) lincomepop y1995, vce(cluster state) fe
```

Coefficient on `lavgprs`: **-1.27** ($p < 0.001$)

Best Elasticity Estimate

When price goes up by 1%, sales go down by **1.3%**. Demand is elastic, but not terribly so. The CI of (-1.53, -1.00) barely excludes -1, so we can statistically reject that demand is *inelastic*.

Comparing All Specifications

	(1) OLS	(2) 2SLS	(3) 2SLS+controls	(4) State FE	(5) State+Year FE	(6) 2SLS Panel	(7) 2SLS Log
Price	-0.385***	-0.421***	-0.687***	-0.355***	-0.416***	-0.409***	
log(price)							-1.269***
Income			2.816***	0.232	-1.642*	-1.681**	
State FE				Yes	Yes	Yes	Yes
Year FE					Yes	Yes	Yes
IV		Yes	Yes			Yes	Yes
N	96	96	96	96	96	96	96

What Changes Across Specifications?

Notice that the price coefficient changes as we add controls and instruments. The preferred specification (column 7) combines all: fixed effects, IV, controls, and logarithms.

Applied Examples

Example 1: Returns to Schooling

$$\log(wage_i) = \beta_0 + \beta_1 schooling_i + \delta V_i + u_i$$

Data show that people who attend college earn high wages

We want to estimate the **causal** effect

OLS can't distinguish whether high wages are due to the causal benefit of schooling or because people who attend college would be able workers regardless

- Innate ability in u

At the extreme: college might just be a way to **signal** to employers a student's innate ability — credentials how innately smart you are

Instrument: Quarter of Birth (Angrist and Krueger, 1991)

Many states do not let you drop out of school until age 16 (some places 17)

High school students turn 16 at different times during the year

Children born **earlier** in the year can drop out **earlier**

So, children born earlier in the year get **less** total schooling

Check the Three Assumptions

1. **Relevant?** Yes — quarter of birth is correlated with schooling attainment through compulsory schooling laws
2. **Exogenous?** Debatable — is quarter of birth correlated with innate ability or other wage determinants? (Bound, Jaeger, and Baker, 1993)
3. **Excludable?** Debatable — does birth quarter directly affect wages, e.g., through age-at-hire effects?

This instrument has been extensively debated — it illustrates how the assumptions can be **challenged** even when the research design seems clever.

Angrist and Krueger (1991): Estimates

TABLE IV
OLS AND TSLS ESTIMATES OF THE RETURN TO EDUCATION FOR MEN BORN 1920–1929: 1970 CENSUS[a]

Independent variable	(1) OLS	(2) TSLS	(3) OLS	(4) TSLS	(5) OLS	(6) TSLS	(7) OLS	(8) TSLS
Years of education	0.0802	0.0769	0.0802	0.1310	0.0701	0.0669	0.0701	0.1007
	(0.0004)	(0.0150)	(0.0004)	(0.0334)	(0.0004)	(0.0151)	(0.0004)	(0.0334)
Race (1 = black)	—	—	—	—	0.2980	−0.3055	−0.2980	−0.2271
					(0.0043)	(0.0353)	(0.0043)	(0.0776)
SMSA (1 = center city)	—	—	—	—	0.1343	0.1362	0.1343	0.1163
					(0.0026)	(0.0092)	(0.0026)	(0.0198)
Married (1 = married)	—	—	—	—	0.2928	0.2941	0.2928	0.2804
					(0.0037)	(0.0072)	(0.0037)	(0.0141)
9 Year-of-birth dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
8 Region of residence dummies	No	No	No	No	Yes	Yes	Yes	Yes
Age	—	—	0.1446	0.1409	—	—	0.1162	0.1170
			(0.0676)	(0.0704)			(0.0652)	(0.0662)
Age-squared	—	—	−0.0015	−0.0014	—	—	−0.0013	−0.0012
			(0.0007)	(0.0008)			(0.0007)	(0.0007)
χ^2 [dof]	—	36.0 [29]	—	25.6 [27]	—	34.2 [29]	—	28.8 [27]

a. Standard errors are in parentheses. Sample size is 247,199. Instruments are a full set of quarter-of-birth times year-of-birth interactions. The sample consists of males born in the United States. The sample is drawn from the State, County, and Neighborhoods 1 percent samples of the 1970 Census (15 percent form). The dependent variable is the log of weekly earnings. Age and age-squared are measured in quarters of years. Each equation also includes an intercept.

OLS and TSLS estimates of returns to education, men born 1920–1929 (1970 Census, $n = 247{,}199$). Instruments: quarter-of-birth × year-of-birth interactions.

Bound, Jaeger, and Baker (1993): Two Critiques

Bound, Jaeger, and Baker (1993) raised two distinct problems with the quarter-of-birth instrument:

Problem 1 — Instrument may not be exogenous: Quarter of birth is empirically correlated with many outcomes beyond schooling:

School attendance, behavioral difficulties, mental health referrals

Performance in reading, writing, and arithmetic

IQ scores and family incomes

If quarter of birth is correlated with ability or other determinants of wages, $\text{Cov}(Z, u) \neq 0$ — the exogeneity assumption fails.

Problem 2 — Weak instruments: With proper age controls, the first-stage F-statistic falls to **1.6** — far below the rule-of-thumb threshold of 10.

> **Combined Effect**
>
> A **weak instrument** combined with even a small $\text{Cov}(Z, u)$ can cause the IV estimate to look similar to **OLS**. The bias “fix” may introduce new bias rather than eliminating the original one.

IV Recipe Card

IV Estimation: Step by Step

Recipe Card: Instrumental Variables

1. Identify the endogeneity problem

Why do you think $\text{Corr}(X, u) \neq 0$?

OVB, simultaneous causality, or measurement error?

2. Propose an instrument Z and argue it is valid

Powerful: $\text{Corr}(Z, X) \neq 0$ (testable)

Exogenous: $\text{Corr}(Z, u) = 0$ (not testable — must argue)

Excluded: Z does not directly cause Y (not testable — must argue)

3. Estimate via 2SLS

```
ivregress 2sls Y (X=Z) controls, robust
```

4. Report and check

Report first-stage F-stat (rule of thumb: > 10)

Discuss threats to exogeneity and exclusion

Compare IV estimates to OLS — if they differ, explain why

Common Pitfalls

Mistakes to avoid

Using a **weak instrument** (F < 10)

Failing to argue **exogeneity**

Forgetting to report the **first stage**

Using manual 2SLS standard errors

Treating IV as a “magic fix” for endogeneity

Good IV practice

- Provide institutional justification for the instrument
- Report the first-stage F-statistic
- Use `ivregress` (not manual two-step)
- Discuss threats to exclusion
- Compare IV to OLS and explain differences

> ⚠ **IV Is Only as Good as the Instrument**
>
> An IV estimate with a bad instrument can be **worse** than OLS. The whole approach depends on finding a convincing instrument — which is why IV papers spend so much time justifying their instruments.

Key Takeaways

Endogeneity ($E(u|X) \neq 0$) makes OLS biased and inconsistent

IV addresses this by finding a variable Z that:

- Affects X (powerful)
- Is uncorrelated with u (exogenous)
- Does not directly affect Y (excluded)

2SLS isolates the exogenous variation in X using Z

First-stage F > 10 is a common rule-of-thumb diagnostic for instrument strength

Exogeneity cannot be tested — it must be argued

IV can be combined with **panel data** and fixed effects

Always use `ivregress` or `xtivreg` — never manual two-step

Final Knowledge Check

Suppose a researcher wants to estimate the effect of class size on student test scores. They propose using school budget per pupil as an instrument for class size. Evaluate: Is this instrument likely to be powerful? Exogenous? Excludable? What might be wrong?